Z-TRANSFORM

Unified Quick Reference

Mathematical Visualisation & Engineering Applications Edition

1 DEFINITION OF THE Z-TRANSFORM

Bilateral (two-sided)

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$

Uses positive and negative indices: causal, anti-causal and two-sided sequences.

Unilateral (one-sided)

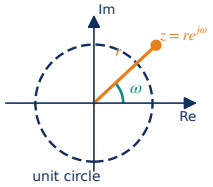
$$X^{+}(z) = \sum_{n=0}^{\infty} x[n]\, z^{-n}$$

Starts at $n=0$: used for difference equations with initial conditions.

Polar form $z = re^{j\omega}$

$$X(re^{j\omega}) = \sum x[n]\, r^{-n} e^{-j\omega n}$$

$r=1$ (unit circle): $X(e^{j\omega})$ is the DTFT,
valid only if $|z|=1$ lies inside the ROC.



2 STANDARD Z-TRANSFORM PAIRS

Sequence $x[n]$	Z-transform $X(z)$	ROC
$\delta[n]$	1	all z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$n\,a^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$u[n]-u[n-N]$	$\frac{1-z^{-N}}{1-z^{-1}}$	$z \neq 0$ *
$\cos(\omega_0 n)\,u[n]$	$\frac{1-\cos\omega_0 z^{-1}}{1-2\cos\omega_0 z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
$\sin(\omega_0 n)\,u[n]$	$\frac{\sin\omega_0 z^{-1}}{1-2\cos\omega_0 z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$

* Finite-duration ROC = whole finite z-plane except uncancelled singularities.

Pulse: $\frac{1-z^{-N}}{1-z^{-1}} = 1 + z^{-1} + \cdots + z^{-(N-1)}$, so the apparent pole at $z=1$ cancels, leaving only $z=0$. $\delta[n]$ has ROC = all z (including 0 and ∞).

$u[n]=1$ **for** $n \geq 0$, **else** 0; $\delta[n]=1$ **for** $n=0$, **else** 0.

3 IMPORTANT PROPERTIES

Linearity

ROC $\supseteq R_1 \cap R_2$

$$a\,x_1[n] + b\,x_2[n] \;\leftrightarrow\; aX_1(z) + bX_2(z)$$

Time shifting

ROC same ($z=0$ or ∞ may change)

$$x[n-n_0] \;\leftrightarrow\; z^{-n_0}X(z)$$

Unilateral delay: $x[n-k] \leftrightarrow z^{-k}X^{+}(z) + \sum_{m=1}^{k} x[-m]\, z^{m-k}$

Unilateral advance: $x[n+1] \leftrightarrow zX^{+}(z) - z\,x[0]$

Multiplication by a^n

ROC $\rightarrow |a|R$

$$a^n x[n] \;\leftrightarrow\; X(z/a)$$

Time reversal

$r_1 < |z| < r_2 \;\rightarrow\; 1/r_2 < |z| < 1/r_1$

$$x[-n] \;\leftrightarrow\; X(z^{-1})$$

Differentiation in the z-domain

ROC unchanged

$$n\,x[n] \;\leftrightarrow\; -z\frac{dX(z)}{dz}$$

Convolution

ROC $\supseteq R_1 \cap R_2$

$$x_1[n] * x_2[n] \;\leftrightarrow\; X_1(z)\,X_2(z)$$

Initial value theorem

causal $x[n]$ only

$$x[0] = \lim_{z\to\infty} X(z)$$

Final value theorem

needs conditions ↓

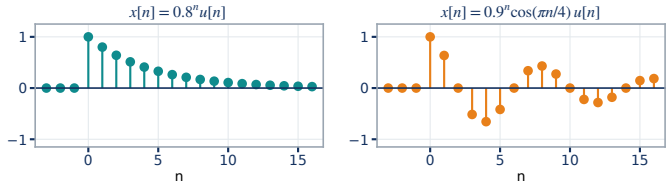
$$\lim_{n\to\infty} x[n] = \lim_{z\to 1}(1-z^{-1})X(z)$$

CAUTION — Final value theorem

Use only if $x[n]$ is causal and every pole of $(1-z^{-1})X(z)$ lies inside $|z|<1$. Example: $x[n]=2^n u[n]$ gives 0 from the formula, but $x[n]\to\infty$. Do not apply it.

DISCRETE-TIME SEQUENCES

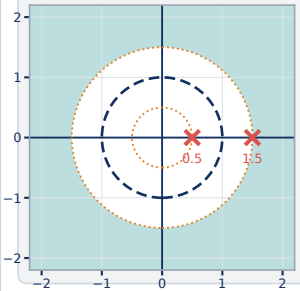
Pole radius sets the decay; pole angle sets the oscillation.



4 REGION OF CONVERGENCE (ROC)

$$\text{ROC} = \{z : \sum_n |x[n]\, z^{-n}| < \infty\}$$

- The ROC contains no poles.
- Right-sided (causal): ROC lies outside the outermost pole.
- Left-sided (anti-causal): ROC lies inside the innermost pole.
- Two-sided: ROC is an annulus between poles.
- BIBO-stable ⇔ ROC contains the unit circle.
- Causal rational system: ROC outside the outermost pole.



Right-sided (causal)

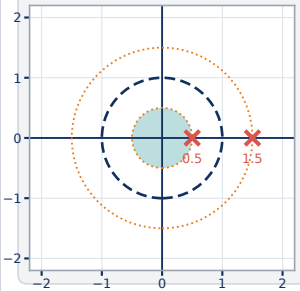
$x[n] = 0.5^n u[n] + 1.5^n u[n]$

ROC: $|z| > 1.5$

outside outermost pole

unit circle ∉ ROC → unstable

poles × at 0.5 and 1.5



Left-sided (anti-causal)

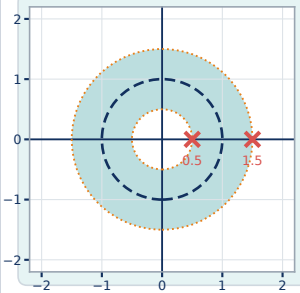
$x[n] = -(0.5^n + 1.5^n)\, u[-n-1]$

ROC: $|z| < 0.5$

inside innermost pole

unit circle ∉ ROC → unstable

poles × at 0.5 and 1.5



Two-sided

$x[n] = 0.5^n u[n] - 1.5^n u[-n-1]$

ROC: $0.5 < |z| < 1.5$

annulus between the poles

unit circle ∈ ROC → stable

poles × at 0.5 and 1.5

pole ○ zero ---- $|z|=1$ ROC

Same $X(z)$ expression, different ROC ⇒ different $x[n]$.
Always state the ROC with the transform.

5 INVERSE Z-TRANSFORM

$$x[n] = \mathcal{Z}^{-1}\{X(z)\} = \frac{1}{2\pi j}\oint_C X(z)\, z^{n-1}\, dz$$

Method-selection table

Method	Suitable situation
Partial fraction decomposition	Rational functions
Power series expansion	Direct coefficient extraction
Long division	Improper rational functions
Residue method	Complex-analysis formulation
Standard transform pairs	Simple expressions
Contour integration	Theoretical inverse transform

6 PARTIAL FRACTION DECOMPOSITION

Distinct linear factors

$$X(z) = \frac{A}{1-az^{-1}} + \frac{B}{1-bz^{-1}}, \quad A = (1-az^{-1})X(z)|_{z=a}$$

Repeated factors

$$X(z) = \frac{A_1}{1-az^{-1}} + \frac{A_2}{(1-az^{-1})^2} + \cdots$$

Pair: $\frac{1}{(1-az^{-1})^2} \leftrightarrow (n+1)a^n u[n], \quad |z|>|a|$

The ROC decides whether each term is right- or left-sided.

Worked example: $X(z) = \frac{1}{1-2z^{-1}}$

ROC $|z|>2$: $x[n] = 2^n u[n]$ (right-sided)

ROC $|z|<2$: $x[n] = -2^n u[-n-1]$ (left-sided)

The expression alone does not fix the inverse; the ROC is essential.

7 DIFFERENCE EQUATIONS ↔ Z-TRANSFORM

$$y[n] + a_1 y[n-1] + \cdots + a_N y[n-N] = b_0 x[n] + \cdots + b_M x[n-M]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{1 + a_1 z^{-1} + \cdots + a_N z^{-N}}$$

(zero initial conditions)

- Numerator roots = zeros; denominator roots = poles.
- Poles and zeros give the system's character and transients.
- The ROC gives convergence, causality and stability.

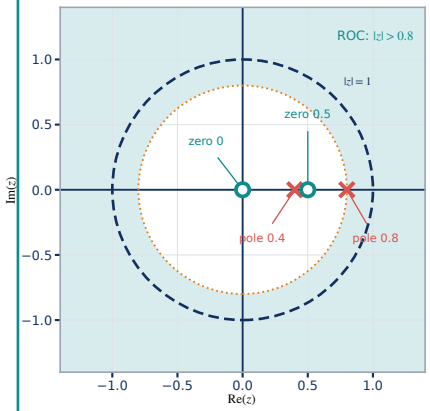
Input $x[n]$ → Difference equation → Z-transform → Transfer function $H(z)$ → Output $y[n]$

Solving with initial conditions:

Apply $X^+(z)$ shift rules → Solve for $Y(z)$ → Partial fractions → Inverse → $y[n]$

8 POLE-ZERO DIAGRAM

$$H(z) = \frac{1-0.5z^{-1}}{1-1.2z^{-1}+0.32z^{-2}} = \frac{z(z-0.5)}{(z-0.4)(z-0.8)}$$



Reading the diagram

Zero: numerator root (0 and 0.5).

Pole: denominator root (0.4 and 0.8).

Unit circle: |z|=1 gives the DTFT / frequency response.

Poles near |z|=1 → slow transients, sharp peaks; poles near 0 → fast decay.

Both poles <1: stable (causal).

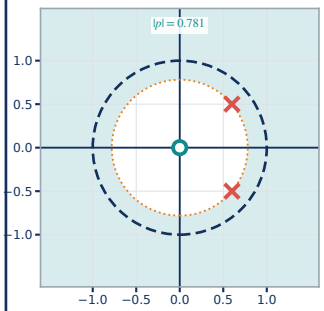
Computed with numpy.roots; the origin zero comes from the padding $z^{N-M} = z$ (M<N). ROC (shaded): $|z|>0.8$.

Geometry: $|H(e^{j\omega})| = \prod|e^{j\omega} - \text{zeros}| / \prod|e^{j\omega} - \text{poles}|$

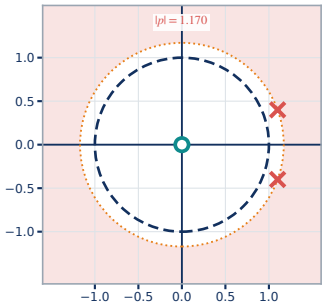
10 STABILITY & FREQUENCY RESPONSE

- BIBO stability (causal rational): all poles strictly inside $|z|=1$.
- Equivalent: the ROC contains the unit circle ($\sum|h[n]|<\infty$).
- Any pole with $|p|\geq 1$ makes a causal system unstable.

Stable: poles inside



Unstable: pole outside



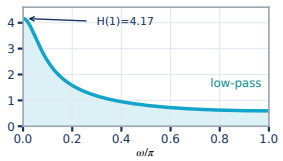
$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$$

Evaluate on the unit circle, valid only when the unit circle lies in the ROC.

$\omega = 2\pi f/f_s$ **(rad/sample)**

$\omega = \pi$ is the Nyquist frequency

$|H(e^{j\omega})|$ of Sec. 8 system



9 ENGINEERING APPLICATIONS

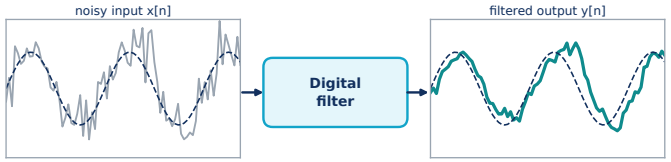
A Digital Signal Processing
- Digital filtering
- Signal reconstruction
- Frequency response
- FIR and IIR filters

B Control Systems
- Discrete-time control
- Digital controllers
- Stability analysis
- System transfer functions

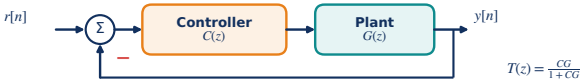
C Difference Equations
- Recurrence relations
- Discrete models
- Population models
- Numerical algorithms

D Digital Communication
- Signal transmission
- Channel modelling
- Discrete-time systems
- Modulation, signal processing

Digital filtering: $y[n] = 0.7\,y[n-1] + 0.3\,x[n]$, $H(z) = \frac{0.3}{1-0.7z^{-1}}$



Digital control loop:



Example: $y[n] - 0.5\,y[n-1] = u[n]$, $y[-1] = 0$

$Y(z) = \frac{1}{(1-z^{-1})(1-0.5z^{-1})} = \frac{2}{1-z^{-1}} - \frac{1}{1-0.5z^{-1}}$

$y[n] = 2 - (0.5)^n$, $n \geq 0$ **(y[0] = 1, y[1] = 1.5; FVT: y[∞] = 2)**

11 QUICK METHOD-SELECTION GUIDE

GIVEN seq. or X(z)

Standard sequence?	Use transform pairs
Rational function?	Partial fractions
Repeated poles?	Repeated-factor decomposition
Difference equation?	Apply Z-transform + initial cond.
Need stability?	Examine poles and ROC
Need frequency response?	Evaluate on the unit circle

12 COMMON ERRORS

- ✗ Ignoring the ROC
- ✗ Wrong shift power z^{-n_0}
- ✗ Wrong partial-fraction constants
- ✗ Poles inside ⇒ stable, ignoring ROC
- ✗ Mixing unilateral & bilateral
- ✗ Dropping initial conditions
- ✗ FVT used without pole check
- ✗ Confusing f with $\omega = 2\pi f/f_s$

Sequence → Z-Transform → ROC → Inverse → Pole-Zero → Stability